



RISK MANAGEMENT PLAN
LITPATH AI: SMART PATHFINDER FOR THESES AND DISSERTATIONS

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INTRODUCTION

This section explains why risks exist and highlights the purpose and importance of the risk management plan. It provides a general description of why risk management is essential to effectively managing a project and describes what is needed before risk management can begin.

As with any technology project involving AI integration, and third-party API dependencies, the project operates within an environment of inherent uncertainty. The purpose of this Risk Management Plan is to establish the framework by which the LitPath AI project team will identify, assess, and develop strategies to mitigate or avoid risks that may threaten the project's objectives, deliverables, and deadlines.

LitPath AI is considered a medium-to-high risk project with an overall risk score of 15.67 on a scale from 1 to 25, derived from the average of all 15 identified risk scores. A risk score below 5 is a low-risk project; a score between 5 and 10 is a medium risk project, and a score above 10 is a high-risk project.

Before risk management activities formally began, it was imperative that a foundation of structured project information be established. The following project elements were completed and defined prior to developing this Risk Management Plan:

- Define work scope, schedule, resources, and cost elements through
 - Statement of Work
 - Scope Management Plan
 - Schedule Management Plan
 - Cost Management Plan
- Define minimum and maximum baseline thresholds
 - Schedule
 - Resources
 - Cost
- Baseline reporting requirements
 - Format
 - Frequency of distribution
 - Distribution list
- Define Risk Management Roles and Responsibilities
 - Project Manager chairs the risk assessment meetings





- Project team participates in risk assessment meetings and members serve as meeting recorder and timekeeper
- Key stakeholders participate in risk assessment meetings

TOP THREE RISKS

It is important to explicitly state the top three risks to the project in the Risk Management Plan. This will make management aware of the top risks for the project and the nature of the risks.

The top three high probability and high impact risks to the LitPath AI project are:

Gemini AI API Unavailability or Rate Limiting

Due to the project's dependency on the Gemini AI API for semantic query understanding, AI-generated overviews, and result re-ranking, any extended service outage, rate limiting, or unexpected deprecation of the API would render the system's core AI search functionality inoperable, directly threatening the project's primary deliverable. The Project Manager will mitigate this risk by implementing response caching for frequent queries, monitoring Gemini AI service status continuously, and coordinating with DOST-STII to ensure API cost coverage is in place before production deployment.

Search Relevance Optimization

Due to the complexity of semantic search across specialized academic literature, the system's RAG pipeline requires continuous refinement to achieve the desired relevance benchmarks. Failure to resolve low-relevance query patterns prior to formal usability testing may result in rework that affects deployment timelines. To mitigate this risk, the Project Manager and Technical Lead will conduct iterative internal relevance evaluations using predefined test queries, continuously refine LLM prompts, and perform internal pre-testing before formal usability testing commences.

LLM Hallucination

Due to the probabilistic nature of large language models, the Gemini AI API may generate inaccurate or misleading outputs that could affect the system's reliability as an academic research tool and risk rejection during acceptance testing. To mitigate this risk, the Project Manager and QA Lead will implement source-grounded prompting, display citations with AI-generated outputs, and conduct hallucination testing using verified dataset entries during the QA cycle.



RISK MANAGEMENT APPROACH

This section provides a general description for the approach taken to identify and manage the risks associated with the project. It should be a short paragraph or two summarizing the approach to risk management on this project.

The LitPath AI project follows a structured risk management approach aligned with its sprint-based development. Risk identification was conducted through a single formal risk assessment meeting held during the project planning phase. The HecTech team identified, scored, and prioritized risks across Technical, Schedule, Resource, External, and Compliance categories during the assessment meeting, with high-impact risks assigned to Risk Managers for active mitigation. These risks are monitored and reported during bi-weekly sprint reviews via Microsoft Teams when relevant to the current sprint window, while the Project Manager continuously updates the Risk Register as risks evolve. Upon project completion, the overall effectiveness of the risk management process is evaluated, with lessons learned documented and archived to support future HecTech projects.

RISK IDENTIFICATION

This section explains the process by which the risks associated with this project were identified. It should describe the method(s) for how the project team identified risks, the format in which risks are recorded, and the forum in which this process was conducted. Typical methods of identifying risks are expert interview, review historical information from similar projects and conducting a risk assessment meeting with the project team and key stakeholders.

Risk identification for the LitPath AI project was conducted through a risk assessment meeting held during the project planning phase, with supplementary input gathered through an expert interview with the Project Clients and a review of similar projects. The Project Manager chaired the risk assessment meeting.

Expert Interview

One expert interview was conducted for the LitPath AI project. It was held with the DOST-STII client representatives to identify operational, compliance, and deployment risks from the client's perspective including data privacy requirements under the Data Privacy Act of 2012 (RA 10173), server hosting constraints, and AI/LLM API cost coverage responsibilities. Risks identified and resolved through this interview, such as adjusting the authentication architecture to comply with RA 10173, were addressed directly in the system design. Remaining risks were logged in the Risk Register.

Risk Assessment Meeting



A risk assessment meeting was held with all four HecTech team members and attended by the Project Clients. The Crawford Slip method was used, giving each participant 10 minutes to independently record as many risks as possible on notepads. The Project Manager then facilitated a group discussion in which risks were read aloud, consolidated, and categorized. Identified risks were added to the Risk Register.

Historical Review of Similar Projects

The project team reviewed post-project documentation and known failure patterns from comparable AI-powered academic search and web application projects to identify common risks, including AI API reliability issues, last-mile deployment failures, user acceptance shortfalls, and scope creep introduced by client feedback during refinement phases. These historical insights informed both the identification process and the development of mitigation strategies, particularly for the Gemini AI dependency risk and the usability testing acceptance threshold risk.

RISK QUALIFICATION AND PRIORITIZATION

Once risks are identified it is important to determine the probability and impact of each risk in order to allow the project manager to prioritize the risk avoidance and mitigation strategy. Risks which are more likely to occur and have a significant impact on the project will be the highest priority risks while those which are more unlikely or have a low impact will be a much lower priority. This is usually done with a probability – impact matrix. This section explains risks were qualified and prioritized for this project. For more information on how to qualify and prioritize risks refer to our ***Risk Assessment Meeting Guide***.

Once risks were identified, a probability and impact factor was assigned to each risk to allow the Project Manager to prioritize mitigation and avoidance strategies. Risks that are more likely to occur and have a significant impact on the project's schedule, cost, scope, or acceptance criteria were classified as the highest priority, while those that are less likely or have limited impact were assigned a lower priority.

Each risk was assigned a probability rating from 1 to 5 and an impact rating from 1 to 5, where 1 represents the lowest likelihood or negligible effect, and 5 represents a near-certain occurrence or a severe effect that directly blocks project delivery. The resulting risk score calculated as probability multiplied by impact falls on a scale from 1 to 25 and is used to categorize risks as Low (1–4), Medium (5–9), High (10–19), or Critical (20–25).

Probability and impact ratings were determined through a structured voting process conducted during the risk assessment meeting. For each identified risk, all four HecTech team members independently assigned a probability rating and an impact rating based on the following defined criteria:



Probability Rating Criteria:

- **1 — Very Low:** The risk event is highly unlikely with no historical precedent in the project context and existing controls making occurrence improbable.
- **2 — Low:** The risk event is unlikely but possible with limited exposure and conditions largely within the team's control.
- **3 — Moderate:** The risk event is possible due to external dependencies or known project constraints outside the team's direct control.
- **4 — High:** The risk event is likely because of strong dependence on external factors or recurring conditions that increase probability.
- **5 — Very High:** The risk event is very likely or near certain due to critical dependencies and the absence of effective mitigation measures.

Impact Rating Criteria:

- **1 — Very Low:** Negligible effect on project objectives with no significant impact on schedule, cost, or sprint capacity.
- **2 — Low:** Minor effect that can be resolved within the sprint cycle with minimal schedule adjustment.
- **3 — Moderate:** Noticeable effect on schedule, cost, or quality requiring corrective action during the sprint cycle.
- **4 — High:** Significant effect that threatens a project milestone or acceptance criterion and requires escalation to the Project Sponsor.
- **5 — Critical:** Severe effect that directly prevents project delivery, formal acceptance, or deployment.

After all team members submitted their ratings, the scores were averaged to determine the final probability and impact values for each risk. The overall risk score was then calculated by multiplying the average ratings, resulting in a score from 1 to 25. This method reflects the collective assessment of the project team and allows clearer differentiation between risks through decimal-based scoring. Individual ratings, averaged values, and computed scores are documented in the Voting Table.

The Project Manager used a probability-impact matrix to position each risk according to its assessed ratings. The matrix applied color coding to indicate severity levels namely green for Low, yellow for Medium, orange for High, and red for Critical risks. After all risks were plotted, the



results were documented and the team proceeded to mitigation and avoidance planning for priority risks.

RISK MONITORING

This section should discuss how the risks in the project will be actively monitored. One effective way to monitor project risks is to add those risks with the highest scores to the project schedule with an assigned risk manager. This allows the project manager to see when these risks need to be monitored more closely and when to expect the risk manager to provide status updates at the bi-weekly project team meetings. The key to risk monitoring is to ensure that it is continuous throughout the life of the project and includes the identification of trigger conditions for each risk and thorough documentation of the process.

Risk monitoring for the LitPath AI project is a continuous process embedded in bi-weekly sprint reviews managed through Microsoft Teams, where all priority risks are tracked with assigned Risk Manager responsible for monitoring status and trigger conditions within their occurrence window. During each review, only active risks relevant to the current sprint are discussed to maintain efficiency, with updates covering status (open, triggered, closed), observed triggers, and response actions taken. The Project Manager then updates the Risk Register and escalates significant issues to the Project Clients via Viber while predefined trigger conditions such as API latency thresholds or sustained drops in sprint velocity guide for timely responses.

RISK MITIGATION AND AVOIDANCE

Once risks have been qualified, the team must determine how to address those risks which have the greatest potential probability and impact on the project. This section explains the considerations which must be made and the options available to the project manager in managing these risks.

The Project Manager, with the full participation of the HecTech development team, has developed a specific avoidance or mitigation response for each risk identified and prioritized in the Risk Register. As additional risks are identified during subsequent sprints, they will be scored, assigned a Risk Manager, and logged into the Risk Register with a corresponding response strategy.

All risk responses for LitPath AI will be managed and controlled within the constraints of time, scope, and cost. Scope and time are the firm constraints for this project. Cost is the flexible constraint of last resort like if a critical risk cannot be mitigated within the existing sprint budget and the PHP 40,000.00 contingency reserve, the Project Manager may request additional funding from the Project Sponsor to cover supplementary resources, extended cloud hosting, or additional AI/LLM API usage subject to a formal Change Request and Sponsor approval. The PHP



4,000.00 Management Reserve held by the Project Sponsor may also be drawn upon in extreme cases, as permitted under the Cost Management Plan.

RISK REGISTER

Every project must maintain a risk register in order to track risks and associated mitigation strategies. This section describes the risk register criteria as well as where the risk register is maintained and how these risks are tracked in the project schedule.

The LitPath AI Risk Register is the central log for all 15 identified risks. For each risk, the register captures: risk title, risk description, category (Technical, Schedule, Resource, External, Compliance, or Cost), averaged probability rating (1–5), averaged impact rating (1–5), overall risk score (1–25), risk ranking, assigned Risk Manager, mitigation or avoidance strategy, and trigger conditions.

The Risk Register was created during the project risk assessment meeting facilitated by the Project Manager. Team members independently assigned probability and impact ratings, which were averaged to produce the final scores and rankings. Prior to each risk's anticipated occurrence period, the Project Manager confirms that mitigation measures and trigger monitoring are active. Assigned Risk Managers then provide status updates during bi-weekly sprint reviews within the relevant timeframe.

The Risk Register is maintained as a living document updated after each bi-weekly review meeting and upon any trigger event and is accessible to all team members and the Project Sponsor via the project's shared document repository on SharePoint and Google Drive. The Risk Register is maintained as Appendix A to this Risk Management Plan.



SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Date: April 30, 2026

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

APPENDIX A

Risk Register: 02 Project Planning 08 Risk-Register.xlsx

Risk Identification			Qualitative Rating				Risk Response		
Risk Title	Risk Description	Risk Category	Probability	Impact	Risk Score	Risk Ranking	Risk Response	Trigger	Risk Owner
API Unavailability	Due to LRP/Path AI's dependency on the Gemini AI API for semantic query processing and AI-generated overviews, an extended service outage, rate limiting, or API deprecation by Google could disable the system's core search functionality and prevent the project from meeting its primary acceptance criteria.	Technical / External	4	5	20	1	Implement response caching for high-frequency queries, establish a fallback degraded-mode search using ChromaDB vector similarity without Gemini AI; coordinate with DOST-STII to confirm API cost coverage before production deployment.	Gemini AI API returns persistent errors, exceeds rate limits, or announces deprecation during development or testing phases.	Charjoy Cempron
Search Relevance Failure	Due to the complexity of semantic search over specialized academic literature, the system's RAG pipeline requires continuous iterative refinement to produce results that consistently meet the team's internal relevance benchmarks. Failure to address low-relevance query patterns before formal usability testing could result in rework cycles that consume sprint capacity originally reserved for deployment activities.	Technical	3.25	5	16.25	2	Conduct iterative internal relevance evaluation throughout development using a defined test query set; refine LLM prompts continuously; perform pre-usability relevance testing at least two weeks before formal usability testing begins.	Internal relevance testing reveals consistent failures across the defined test query set, or early informal user feedback surfaces recurring irrelevant results.	Charjoy Cempron
LLM Hallucination	Due to the inherent probabilistic nature of large language models, the Gemini AI API may generate inaccurate, fabricated, or misleading summaries, citations, or search overviews that appear credible to users, undermining the system's reliability as an academic research tool and potentially causing DOST-STII to reject the deliverable during acceptance testing.	Technical / AI	3	5	15	3	Implement source-grounded prompting that constrains Gemini AI responses strictly to retrieved document content; display source citations alongside every AI-generated output; conduct targeted hallucination testing as part of the QA cycle using known dataset entries to verify output accuracy.	QA testing identifies AI-generated summaries or citations that do not correspond to actual content in the indexed dataset, or usability testing participants report misleading or fabricated search results.	Joshua Sta Ana
API Latency / Slow Response	Due to LRP/Path AI's reliance on external API calls to Gemini AI for query processing and result generation, slow API response times under load or degraded network conditions could cause search query response times to exceed the ten seconds threshold, failing the system's performance acceptance criterion even when the API is technically available.	Technical / External	3.5	4	14	4	Implement timeout handling with defined maximum wait thresholds; use asynchronous API calls to prevent blocking; cache responses for frequent or repeated queries; display partial results while full AI overview loads to maintain perceived performance.	API response times consistently exceed 10 seconds during internal load testing, or usability testing participants report noticeable delays between query submission and results display.	Charjoy Cempron
Dataset Provision Delay	Due to the project's reliance on DOST-STII to provide a representative sample dataset of theses and dissertations, any delay or inadequacy in the dataset handover could prevent the team from completing RAG pipeline validation, relevance testing, and system acceptance within the project timeline.	External	3.25	4	13	5	Confirm dataset delivery schedule with DOST-STII at the start of each major testing phase; request a preliminary sample early to identify format or volume issues; define minimum dataset requirements in writing with the Project Sponsor.	DOST-STII has not provided the required dataset within two weeks of the agreed handover date, or the dataset is insufficient in volume or metadata quality for testing.	Tracee Tomon
Cloud Hosting Delay	Due to the planned transition from the local Promox development environment to a cloud hosting environment managed by DOST-STII for final production deployment, delays in confirming the hosting setup or cost approvals could prevent the system from going live by the June-July 2026 deadline.	External / Schedule	3	4	12	6	Initiate cloud hosting discussions with DOST-STII no later than May 2026; document hosting requirements in the Statement of Work; maintain the Promox server as a fallback demo environment; escalate any delays immediately to the Project Sponsor.	DOST-STII has not confirmed the production hosting environment or cost coverage by May 15, 2026, or the cloud environment setup encounters technical blockers during pre-deployment testing.	Tracee Tomon



PROJECT MANAGEMENT DOCS

ProjectManagementDocs.com

Database Inconsistency	Due to the system's use of Supabase (PostgreSQL) for relational data and ChromaDB for vector embeddings, a failure in data consistency between the two systems could corrupt search results, break the bookmarking feature, or prevent accurate citation generation.	Technical	2.75	4	11	7	Implement automated consistency checks between Supabase and ChromaDB after each data ingestion cycle; maintain backup storage on the HDD per the system design; establish a documented data recovery procedure; test database integrity in every sprint QA cycle.	Search results return mismatched or duplicate records; bookmarked items fail to load; ChromaDB embeddings become out of sync with Supabase document records.	Joshua Sta Ana
Data Loss in Ingestion	Due to the complexity of the data ingestion and embedding generation pipeline, failures during batch processing — such as interrupted ingestion jobs, malformed document records, or embedding errors — may result in theses or dissertations being silently omitted from the indexed dataset, causing incomplete search results without any visible system error.	Technical	3	3.25	9.75	8	Implement comprehensive ingestion logging with record-level success and failure tracking; add post-ingestion validation that compares source document count against indexed record count; implement retry mechanisms for failed embedding generations; alert the team if record count discrepancies exceed a defined threshold.	Post-ingestion validation reveals a mismatch between the number of source documents provided and the number of records successfully indexed in ChromaDB or Supabase.	Charloey Cempron
Scope Creep	Due to DOST-STII's ability to submit feedback and change requests during the Feature Refinement and usability testing phases, uncontrolled additions beyond the agreed Statement of Work could consume the team's limited sprint capacity and jeopardize the June-July 2026 deadline.	Scope	3	3	9	9	Enforce the formal scope change control process for all client-initiated change requests; document and confirm all feature clarifications in writing before development; require Project Sponsor approval for any changes affecting schedule or budget.	DOST-STII requests a new feature or significant modification outside the defined scope in the Statement of Work during the testing or deployment phase.	Tracie Yomon
Data Privacy Non-Compliance	Due to the Data Privacy Act of 2012 (RA 10173) requirements applicable to LUPATH AI, any failure to implement proper data handling for user accounts, search history, and admin access could result in a compliance violation that blocks formal project acceptance or requires significant post-deployment rework.	Compliance	2	4.25	8.5	10	Ensure all user authentication uses secure login with role-based access control and session management with secure tokens; keep public search anonymous; enforce HTTPS/SSL on all connections; conduct a compliance review led by the Security Lead before usability testing.	Security audit identifies unprotected user data, unsecured API endpoints, or missing authentication controls during internal testing.	Joshua Sta Ana
Dependency / Library Compatibility	Due to the project's reliance on a Python-based AI stack including SentenceTransformers, ChromaDB, LangChain-type utilities, and Supabase client libraries, updates or breaking changes to any of these dependencies could introduce runtime errors, degrade system performance, or require unplanned refactoring that consumes sprint capacity.	Technical	2.25	3.75	8.44	11	Pin all dependency versions in the project's requirements file and lock file; test all updates in the staging environment before applying to the development or production environment; monitor dependency changelogs for breaking changes before each sprint.	System errors occur after a library update or environment change, or a dependency announces a breaking change in an upcoming release that affects a core system component.	Charloey Cempron
Key Resource Unavailability	Due to the four-member HecTech team having no external outsourcing capacity, the unexpected unavailability of a key technical resource due to illness, personal emergency, or academic withdrawal could significantly disrupt sprint delivery and system integration milestones.	Resource	2	4	8	12	Maintain up-to-date technical documentation for all system components; cross-train team members on critical modules within sprint capacity; redistribute tasks among remaining members in the event of a temporary absence; escalate to Project Sponsor if absence exceeds one sprint.	A team member is unable to perform assigned sprint tasks for more than three consecutive working days due to illness, personal emergency, or academic withdrawal.	Tracie Yomon
Search Performance Degradation	Due to the system's use of SentenceTransformers for embedding generation and ChromaDB for vector search, performance degradation under concurrent user load or large dataset volume could cause search query response times to exceed the required ten seconds threshold, failing the system performance acceptance criterion.	Technical	2.25	3	6.75	13	Conduct load and performance testing during the Usability Testing phase; optimize ChromaDB indexing and embedding retrieval pipelines; implement query result caching for high-frequency searches; address all performance bottlenecks before final deployment.	Internal load testing reveals average search response times exceeding ten seconds under simulated concurrent user conditions; or usability testing participants report noticeable delays.	Charloey Cempron
Local Server Hardware Failure	Due to the project's use of a locally hosted Proxmox server for development and internal testing, hardware failure such as SSD failure or power interruption could result in loss of code, database records, or configuration that delays sprint delivery.	Technical	2	3.25	6.5	14	Maintain all source code in a remote Git repository with regular commits after every work session; back up the Supabase database and ChromaDB embeddings to the 500 GB HDD; document all server configuration settings to enable rapid environment recovery.	The Proxmox server becomes inaccessible for more than 24 hours, or a team member reports data loss or file corruption on the local development environment.	Joshua Sta Ana
API Cost Overrun	Due to the project's use of the Gemini AI API without a formal written cost commitment from DOST-STII, any unexpected spike in API usage during testing or deployment could result in costs that exceed the project's contingency reserve and interrupt system functionality.	Cost / External	2	3	6	15	Confirm DOST-STII's formal commitment to cover all cloud hosting and AILLM API costs in writing before transitioning to production; monitor API usage costs throughout testing; report any projected overruns to the Project Sponsor immediately.	DOST-STII requests a delay in API budget commitment, or Gemini AI API usage costs during testing exceed the informally estimated monthly threshold.	Tracie Yomon